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United States Patent	12403355
Kind Code	B2
Date of Patent	September 02, 2025
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Core training device and method

Abstract

The present invention provides an exercise device (**100**) for exercising core muscles of a user's torso. The exercise device comprises a frame (**110**), configured to be supported by the user and a radially outwardly facing channel (**130**; FIG. 6), the channel (**130**) configured to receive a resistance member. The frame defines an aperture (**112**) configured to receive a torso of the user and the device (**100**) is configured to provide a resistance force against which the user can exercise when the device (**100**) is in use. Also claimed is a kit of parts comprising the device (**100**) and a resistance member, and a method of exercising core muscles of one's torso.

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Appl. No.:	17/914625
Filed (or PCT Filed):	March 23, 2021
PCT No.:	PCT/GB2021/050705
PCT Pub. No.:	WO2021/191595
PCT Pub. Date:	September 30, 2021

Prior Publication Data

Document Identifier	Publication Date
US 20230123007 A1	Apr. 20, 2023

Foreign Application Priority Data

GB	2004311	Mar. 25, 2020
GB	2009354	Jun. 18, 2020

GB	2010343	Jul. 06, 2020
GB	2012040	Aug. 03, 2020
GB	2017336	Nov. 02, 2020
GB	2017862	Nov. 12, 2020
GB	2019070	Dec. 03, 2020

Publication Classification

Int. Cl.: A63B23/02 (20060101); A63B21/00 (20060101)

U.S. Cl.:

CPC A63B23/0211 (20130101); A63B21/154 (20130101); A63B21/4007 (20151001); A63B21/4035 (20151001); A63B21/4047 (20151001); A63B2225/685 (20130101)

Field of Classification Search

CPC: A63B (23/0211); A63B (21/154); A63B (21/4007); A63B (21/4035); A63B (21/4047); A63B (2225/685); A63B (2071/0063); A63B (21/00061); A63B (21/00065); A63B (21/0552); A63B (21/0557); A63B (21/169); A63B (21/222); A63B (21/4005); A63B (21/4017); A63B (21/4039); A63B (21/4049); A63B (23/03525); A63B (23/0355); A63B (21/0004)

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Background/Summary

TECHNICAL FIELD

(1) The disclosure relates generally to exercise devices. More particularly, the disclosure relates to portable exercise devices, and more specifically, the disclosure relates to portable exercise devices for training the core muscles of the body, and to methods for doing so.

BACKGROUND

(2) The core muscles of the torso are essential for most activities involving movement. Core muscle strength is required for a variety of sports and particularly for sports in which a club, bat or racquet is swung around the body.

(3) Many exercise devices exist which are suitable for training the core muscles. One example is a rowing machine, with which a user simulates the motion of rowing whilst pulling a handle which provides resistance. Whilst devices such as rowing machines provide suitable pieces of equipment with which to exercise, they are expensive and take up a large amount of space, making them unsuitable for many people. Additionally, some people may travel frequently and therefore they may require equipment which is portable.

(4) If a user adopts a particular (bad) posture during exercise, he or she may become injured and may damage his or her muscles, which can in turn cause the user to experience pain. After sustaining an injury it is more difficult for a user to exercise, even if different muscles are being trained to those which are injured. As such, it is important for a user to maintain a predetermined, correct, posture which reduces his or her risk of becoming injured during exercise. People who have not exercised regularly in the past may be unaware of how to position themselves if he or she wishes to avoid injury, or indeed how to position themselves to get the most effective muscle development. Some of the exercise devices which are currently available require the user to operate the device in a very specific way, which the user may be unaware of.

(5) Low intensity exercise can be beneficial for a wide variety of people with different needs. For example, for a person who is recovering from a previous injury and is undergoing rehabilitation, high intensity exercise can cause the injury to reoccur. As such, he or she may choose to start off with low intensity exercise initially in order to prevent this. Similarly, a person who is currently injured may want to maintain his or her general fitness levels by continuing to perform low intensity exercise which will not affect the injured part of his or her body. It is therefore beneficial to be able to exercise specific muscles or muscle groups, in a controlled predictable way.

(6) Some exercise devices which are currently available only operate within certain limits of resistance to movement and do not provide a big enough variation in difficulty to accommodate for the people of different needs. For example, some devices which operate using weights may not provide a level of exercise which accessible to people who are new to exercising certain muscles or

muscle groups. In contrast, certain devices only operate at a level which is not demanding enough for an experienced user to be able to exercise properly.

(7) An exercise device for training the core muscles that addresses at least some of the issues described above is desired.

SUMMARY OF INVENTION

(8) According to an aspect of the invention there is provided an exercise device for exercising core muscles of a user's torso. The exercise device comprises a frame, configured to be supported by the user and a radially outwardly facing channel, the channel configured to receive a resistance member. The frame defines an aperture configured to receive a torso of the user and the device is configured to provide a resistance force against which the user can exercise when the device is in use.

(9) The aperture may be partly defined by the frame and partly defined by another component such as a strap or support. The aperture may be defined as a void into which a user can place his or her torso. The aperture may be an open aperture, in that the frame does not completely encompass a user when the device is in use.

(10) The frame may have a curved bow shape. The frame may have a convex outer surface that defines the channel. The frame may have a concave and/or padded inner portion adapted to abut or be closely spaced from the user's torso. The aperture may be about 180 degrees and the bow may extend about 180 degrees about the torso/aperture when the device is in use.

(11) The channel may be discontinuous. The device may comprise a first frame section and a plurality of second frame sections extending radially outwardly from the first frame section. Each of the second frame sections may comprise part of the channel. The device may comprise an upper rail and a lower rail. Each of the upper rail and lower rail may be coupled to a frame member of the device by a joint at a plurality of points along the length of the frame member. The rails and joints may form a discontinuous channel. The upper and rails may militate against a resistance member from becoming dislodged from the channel in use.

(12) The frame may be formed from an outer curve and an inner curve that meet at junctions. The junctions may be at opposite ends of the frame. The frame may comprise one or more struts that join the inner and outer curves. The frame may line in a common plane. In other words, the strut or struts and inner and outer curves may lie on a single, common plane.

(13) The frame may comprise a maximum radius in the range of 200-1000 mm, 200-500 mm, or 300-400 mm. The frame may comprise a maximum thickness, measured substantially perpendicularly to a radius of the frame, in the range of 20-100 mm, 20-50 mm, or 30-50 mm. A maximum radius of the frame may be in the range of 2-50, 2-20, 2-10 or 5-10 times a maximum thickness of the frame, measured substantially perpendicularly to the maximum radius of the frame. The thickness of the frame excludes any handles which may project away from a generally planar nature of the frame.

(14) At least part of the frame, such as the inner and/or outer curve, may be constructed from a curved box section, for example a metal curved box section. A cross-section of the curved box section may be square, or any other suitable polygonal shape such as rectangular, triangular or circular. Where the cross-section of the curved box section is square, the height and width of the square cross-section may be in the range of 20-100 mm, 20-50 mm, or 30-50 mm.

(15) The inner curve may be flexible. A flexible inner curve may allow the device to flex to various body shapes of prospective users.

(16) The device may be substantially symmetrical across a sagittal plane and/or a transverse plane of the user's body when the device is held in a use position by the user.

(17) The curved bow shape may be formed along an arc of 120° or more. The aperture may be formed between a corresponding arc of 240° or less.

(18) The device may comprise a thoracic support adapted in use to engage the front of the user's torso. The device may comprise a lumbar support adapted in use to contact the back of a user. One

or both of the thoracic support and the lumbar support may be padded.

(19) The device may comprise a closure strap that in use extends from the ends of the bow shaped frame behind a user, and may be attached to one or both ends of the bow. One or both of the attachments between the strap and the bow may be releasable attachments. One attachment may be a fixed attachment and the other may be a releasable attachment. The closure strap may be adjustable in length. The closure strap may have the lumbar support on it. The lumbar support may be attached to the closure strap or formed within it.

(20) The resistance member may be attachable or attached to one end of the bow.

(21) The resistance member may be attachable or attached to a centre point of the bow.

(22) A pair of resistance members may be provided, or more than two.

(23) The resistance force may be provided as a tangential force. Using a tangential force can provide the user with a uniform and/or controllable force. This can help a user with form and technique and prevent a user from 'cheating' by varying arm position or resistance during an exercise movement.

(24) The resistance force may be configurable by the user. This can allow the user to increase or decrease intensity of exercises accordingly. It can also allow the device to be used by both physically stronger and weaker users.

(25) The exercise device may be light weight and may have a mass of less than 10 kg. The exercise device may be readily picked up by a user and be transportable.

(26) The aperture may have a diameter of between 30 cm and 130 cm. The aperture may be sized to fit a range of human torsos. Various iterations of the device may have varying aperture sizes in order to be used by different target users.

(27) The channel may be situated at, proximal to, or radially inwardly displaced from a perimeter of the frame.

(28) The channel may have a depth that is about 50% as deep as the diameter of the resistance member, or deeper; or about as deep as the diameter of the resistance member or deeper; or of at least twice the diameter of the resistance member, and preferably at least four times or six times.

(29) Having a channel depth greater than a diameter of the resistance member reduces the chances of the resistance member being displaced from the channel. A deeper channel can further reduce the chances of displacement.

(30) The frame may lie on a plane and the channel may extend at one or more points in a direction transverse to the plane. Extending the channel may broaden or deepen or both and broaden and deepen the channel. Broadening and/or deepening the channel can mitigate the resistance member coming unseated from the channel. The broadening may also militate against any rebound of the resistance member should it become detached. This may prevent against the resistance member 'twanging' and potentially striking the user. The extension or extensions to the channel may comprise 'anti-twang' clips or shields. The clips or shields may be releasable components that are releasably attachable to the frame.

(31) The 'anti-twang' clips or shields may extend 5 cm, 10 cm or more transverse to the plane of the frame and may optionally extend substantially equidistantly above and below the plane of the frame.

(32) The aperture may have a central axis that is collinear with a central axis of the frame.

(33) Aligning the axes may provide an ergonomically preferable stance to the user and may also provide improved posture and alignment of the users limbs when exercising with the device.

(34) The exercise device may further comprise handles for providing a device supporting means to the user.

(35) The exercise device may comprise multiple sets of handles, and/or handles that are adjustable relative to the frame, to provide differing hand/arm position for the user when the device is in use.

(36) The frame may define one or more slots that face in a downwards direction when the device is in use. The slots may be configured to receive the handles. The slots may be curved slots. The

curved slots may follow the curve of the frame. The handles may be angularly movable within the slots. The handles may be securable at a plurality of positions within the slots. Securing the handles at a position within the slots may comprise locking the handles in place in a position that is appropriate for the user to hold and support the device.

(37) The handles may be positioned on the inner or the outer curve.

(38) Providing multiple handles, or adjustable handles, may allow different users to use the device (for example users having differing arms spans). It may also provide a user with the option to vary his hand position in order to enact different exercise movements.

(39) The frame may lie on a plane and the handles may extend transversely to the plane. Having the handles perpendicular, or otherwise angled to, a plane of the frame may provide a more ergonomic position for the users hands and/or arms.

(40) The handles may be radially outwardly displaced from the aperture.

(41) The frame may comprise a plurality of releasably attachable sections.

(42) Having releasably attachable sections may provide the device with a means for it to be dismantled for storage and/or transport. It may also provide a means for sections of the device to be swapped or modified. For example a size of the aperture may be modified by providing various inner sections that define different sized apertures in order to fit different user body types and sizes.

(43) The exercise device may comprise an attachment point for attaching the resistance member to. The exercise device may comprise a plurality of attachment points for attaching the resistance member to.

(44) The exercise may comprise an attachment point for attaching the resistance member at a first end of the bow.

(45) The exercise may comprise an attachment point for attaching the resistance member at a second end of the bow.

(46) The exercise may comprise an attachment point for attaching the resistance member to at a centre point of the bow.

(47) A plurality of resistance members may be used with a plurality of attachment points. A plurality of resistance members may be used with a single attachment point. Using multiple attachment points and resistance members may provide a user a means to alter a twisting direction without reconfiguring the device, providing a means to vary exercises and ensure balanced work-out routines.

(48) The attachment point may be a hole (through which the resistance member could be passed and tied or restricted by), a strut (around which the resistance member could be tied or a clip/carabiner or the like of the resistance member could be clipped) or a clip or carabiner or the like (which could attach to a corresponding loop, clip or the like of the resistance member).

(49) When in-use, a portion of the resistance member may be seated in the channel.

(50) The resistance member may be seated in the channel along an arc of 90° or more.

(51) Seating the resistance member may provide a tangential relationship between the exercise device and a resistance point, and therefore a tangential resistance force. The exercise device may comprise the resistance member.

(52) The resistance member may be attached to the frame or another structural part of the device.

(53) The resistance member may comprise a cable (band, strap, rope, or other elongate apparatus) configured to be connected to the frame and further configured to be connected to a resistance point.

(54) The cable may be in the form of a steel rope cable commonly found in commercial gym exercise equipment, or may be of lower gauge. A band may be elasticated.

(55) The resistance member may have a length that is greater than half of a length of a perimeter of the frame. The resistance member may have a length that is greater than 1×, 1.5×, or 2×, or 2.5×, or 3× the length of a perimeter of the frame. This may allow the resistance member to be seated in the channel for a full exercise movement of the user when the user is using the device.

(56) The resistance member may be inextensible. The resistance member may be attached to an external resistance providing apparatus when in use to provide the resistance force.

(57) The resistance member may be extensible. Extension of the resistance member may provide the resistance force.

(58) The device may comprise an indicator to provide a user of the device with information relating to their use of the device. The indicator may comprise a spirit level which enables a user to maintain a given angle of the device relative to the horizontal. The indicator may comprise a rep counter which informs a user of the device of a number of completed repetitions of a particular exercise. The indicator may provide a user of the device with information relating to a force or tension applied to the resistance member. The indicator may comprise an electronic display, such as an LCD screen, or one or more lights, such as an LED, configured to provide a user of the device with information. The indicator may comprise a speaker to provide information to the user in the form of sound. The indicator may comprise an accelerometer, gyroscope or any other suitable sensor or means to determine the motion of the device. The device may comprise mounting and/or attachment means for mounting or attaching an electronic device, such as a mobile phone, tablet or the like, to the device. The device may utilise sensors within the electronic device to determine motion of the device in use. This information may be provided to a user of the device by means of a screen and/or a speaker of the electronic device.

(59) The mounting means may comprise a pocket, slot, housing, or holster for the device—such as a phone. The pocket/holster/receiving slot or housing may in use receive the device/phone and hold it securely and rigidly to the device so that it moves with the device and so that movement of the so-mounted phone/electronic device equates to movement of the exercising device.

(60) In an example use case, the indicator may comprise an LED and a speaker. A user of the device may provide an input to a controller of the device which is indicative of a particular exercise the user wishes to perform. The LED may flash red if the user is performing the exercise incorrectly. The LED may flash green and the speaker may emit a reassuring sound, such as a ‘swoosh’, if the user performs the exercise correctly.

(61) The device may comprise digital control units located in the frame and in the handles. The digital control units may provide a feedback mechanism for the motion of the device which would specifically give the user a report on the use of the device in operation. This reporting for example may include the amount of resistance being applied in operation of the device and the orientation of the device in respect to the angle of use and the path motion of use. The device may be configured to dynamically control the amount of resistance when used with a cable system which supports the appropriate digital technology to provide this capability. The device handles may be configured to control resistance by way of buttons e.g., a “Smart Handle” with Bluetooth functionality giving the user the ability to turn the resistance on/off or increase/decrease resistance.

(62) Another aspect of the invention provides a kit of parts. The kit of parts comprises the exercise device of the preceding aspect, a resistance member and a set of instructions for providing a user with methods of using the device.

(63) Another aspect of the invention provides a method of exercising core muscles of one's torso. The method comprises situating a frame about one's torso and supporting the frame with one's hands, applying a tangential force to the frame, and turning through an angular distance to enact a resistance force against the tangential force.

(64) Another aspect of the invention provides a method of exercising core muscles of one's torso. The method comprises taking the exercise device of the first aspect, attaching one end of a resistance member to the exercise device, attaching another end of the resistance member to a resistance point, putting the device overhead and positioning it around one's torso, positioning oneself so that there is tension in the resistance member, twisting one's torso against the tension so as to exercise the core muscles of one's torso.

(65) Attaching another end of the resistance member to a resistance point may comprise attaching

the resistance member to an exercise machine comprising a weight and pulley system.

(66) According to another aspect of the invention, there is provided an exercise device for exercising a user's core muscles, the exercise device comprising: a frame, wherein the frame defines a central aperture configured to receive a torso or trunk of a user; and an attachment point wherein the attachment point is configured to attach the frame to a resistance member in order to provide a resistance force against which the user can exercise when the device is in use.

(67) The exercise device may be configured such that, in use, the resistance force is provided as a torque about a rotational axis of the frame in response to the twisting or turning his or her torso.

(68) The torque may be a configurable torque. On other words a magnitude of the torque may be adjusted. The direction of the torque can also be reversed so as to exercise difference muscle groups and sides of the user's body.

(69) The torque may be configurable by adjusting a moment of inertia about a central axis of the device, or by changing the resistance of the resistance member, or by changing a resistance provided by a resistance point to which the resistance member is attached (such as a conventional pulley weight type exercise machine).

(70) The device may further comprise fasteners which are configured to attach a first frame section to a second frame section.

(71) The fasteners may be adjustable fasteners which are configured to vary a diameter of an inner frame section, wherein the inner frame section defines the aperture. The fasteners are adjustable through having a length that is variable, such as through a turnbuckle, strap or ratchet system.

(72) The moment of inertia of the exercise device about the central axis of the device may be adjustable.

(73) The frame may comprise receiving means configured to support a resistance member. That is the receiving means can both receive and support the resistance member. The receiving means may be in the form of a channel. The receiving means reduces the likelihood of the resistance member from coming unseated from the exercise device when in use as the resistance member is kept support by the receiving means.

(74) The resistance member may be attached to, or configured to be attached to, the attachment point at a first end of the resistance member and the resistance member may be configured to be attached to an external resistance point at a second end of the resistance member.

(75) The exercise device may comprise multiple attachments points, wherein the multiple attachment points are configured to be attached to the first end of the resistance member. The multiple attachment points may also form the receiving means in that the attachment points also attach a top section of the frame to a bottom section of the frame and the resistance member can be supported against the attachment points when the resistance member is under tension when in use.

(76) According to another aspect of the invention there may be a kit of parts, the kit of parts comprising an exercise device according to the first aspect, and a resistance member.

(77) The device may be configured such that when a resistance member is attached to the frame, the resistance member provides a resistance force. The resistance force may increase and decrease in response to the user twisting the exercise device. The resistance force may be applied in a direction which is substantially tangential to the perimeter of the frame. The resistance force may be applied in a direction such that a component of the force acts in a direction which is tangential to the perimeter of the frame. The resistance force may provide a torque. The torque may be generated in a direction opposite to that in which the user rotates. The user resists (i.e. provides an opposing torque to) this torque as a means of exercise. The user twists against this torque as a means of exercise.

(78) The frame may comprise multiple attachment points. The different attachment points may be used by a user to provide exercise at different intensity levels. The resistance member may be configured to be attached to the different attachment points in order to produce different torques for the user to exercise against. The resistance member may be configured to be attached to the

different attachment points in order to allow the user to exercise different muscle groups.

(79) The frame may be configured such that an axis of rotation of the frame aligns with a central axis of the user during exercise. The frame may be configured to ensure that the user maintains a correct posture whilst exercising. The term correct posture is used throughout this application to refer to a stance during exercise wherein a user minimises his or her chance of sustaining an injury.

(80) The frame of the exercise device may have the shape of a circle. The frame of the exercise device may have the shape of any regular polygon, any ovoid or conical section, or any irregular polygon any other irregular shapes, such as a pear shape.

(81) The frame may comprise an inner diameter and an outer diameter. The inner diameter may be defined by the size of the central aperture of the frame. The outer diameter may define the overall radial extent of the frame.

(82) The inner diameter of the frame may range between 40 cm to 100 cm, or 50 cm to 90 cm, or between 60 cm to 80 cm. The inner diameter of the frame may be adjustable. The inner diameter of the frame may adjustable to enable a variety of different users to fit within the central aperture of the device.

(83) The outer diameter of the frame may range between 70 cm to 120 cm, or between 80 cm to 110 cm, or between 85 cm to 105 cm. The outer diameter of the frame may be no greater than 10 cm wider than an average user's arm span.

(84) The frame of the exercise device may comprise a single frame section. The frame of the exercise device may comprise a plurality of frame sections. The frame sections may be held together via adjustable fasteners. The frame sections may be held together via rigid fasteners. The frame section with the smallest diameter may be referred to as the innermost frame section. The frame section with the largest diameter may be referred to as the outermost frame section.

(85) The frame may comprise a first frame section and a second frame section. A diameter of the first frame section may be smaller than a diameter of the second frame section. The first frame section may define the inner diameter of the frame. The second frame section may define the outer diameter of the frame. The first frame section may be the innermost frame section. The second frame section may be the outermost frame section.

(86) The innermost frame section may be configured to fit proximally around the waist of a user. The term “proximally” in this context means there is a gap between the user's waist and the innermost frame section. The gap between the user and the innermost frame section may range between 0 cm and 30 cm, or between 0 cm and 20 cm, or may range between 0 cm and 10 cm.

(87) The attachment point may be attached to or formed as part of the outermost frame section. The attachment point may be attached to or formed as part of the innermost frame section. Multiple frame sections may have an attachment point attached to or formed as part of them. There may be multiple attachment points on each frame section.

(88) The attachment point may be configured to have the resistance member tied around it. The attachment point may be configured by comprising a projecting rod around which the resistance member can be tied without the subsequent knot slipping off, such as by terminating ends of the rod into the frame. The attachment point may be configured to support a hook which is attached to the resistance member. The attachment point may be configured such that the resistance member can clip on to the frame of the exercise device. The attachment point may be configured to limit the motion of the resistance member from sliding around the frame of the exercise device. The attachment point may be configured to reduce the chance of the resistance member becoming detached from the frame when compared with a device which does not have an attachment point. The attachment point may be configured to have the resistance member attached to it through the provision of corresponding features on the resistance member and the attachment point, such as the attachment point comprise a clip (such as a carabiner) and the resistance member comprise a closed loop, or vice-versa.

(89) A user may selectively connect the resistance member to a particular frame section in order to

select a desired resistance torque to exercise against.

(90) The component of the resistance force which acts in a tangential direction to the perimeter of the frame creates a moment about the centre of rotation of the device. This moment is proportional to the distance between the centre of rotation and the position at which the force is being applied. As such, the attachment points to which the resistance member may be attached provide a greater moment, and hence a greater torque for a user to exercise against, when he or she is positioned further from the centre of rotation of the device. The device enables the resistance member to be attached at differing radii in order to provide differing torques to the user.

(91) The resistance member may apply a rotational force to the frame consistently around the entire perimeter of the frame. The resistance member may apply a rotational force along a continuous section of the frame. The resistance member may apply a rotational force to the innermost frame section. The resistance member may apply a rotation force to the outermost frame section. The resistance member may apply a rotational force to a frame section which does not comprise the attachment point. The resistance member may be attach to the frame via an attachment point on the first frame section and apply a rotational force to the frame via the second frame section or vice versa.

(92) The exercise device may comprise handles. The handles may be positioned on the frame. The exercise device may alternatively comprise detachable handles. The detachable handles may be configured to be attached to or detached from the frame. The term “handles” used throughout the application may now refer to either permanent, or detachable handles. The exercise device may comprise a plurality of handles at varying distances from the centre of the device. Having detachable handles may provide a more compact storage and transport solution as well as allowing a user to position handles according to his or her preference/according to an exercise requirement.

(93) The handles may be positioned on the innermost frame section. The handles may be positioned on the outermost frame section. There may be handles on any of the frame sections. There may be handles positioned on multiple frame sections at the same time. There may be handles positioned at different distances from the centre of rotation of the device. The handles positioned on different frame sections may be configured to enable the user to exercise different muscles or muscle groups. The handles positioned on different frame sections may be configured to enable the user to vary the intensity of exercise he or she experiences. The handles may be positioned at various distances from the centre of the device so that users with different arm spans can reach the handles.

(94) The handles positioned on the outermost frame reduce the force a user needs to apply to rotate the device. The handles positioned on the innermost frame sections increase the force a user needs to apply to rotate the device.

(95) The force which the user may apply via the handles creates a moment about the centre of rotation. The moment is proportional to the distance between the centre of rotation of the device and the position at which the force which creates the moment is applied. As such, the user can apply a greater moment to the device whilst applying the same force by applying the force using handles which are positioned at a greater distance from the centre of rotation of the device.

(96) The exercise device may comprise attachment points at various positions relative to the handles of the device. The attachment points at various positions relative to the handles may be configured to allow a user to exercise different muscles or muscle groups.

(97) The or each handle of any of the above described embodiments may comprise two upright sections and a transverse section extending between the upright sections. The or each handle may be attached to the frame at a first end of each upright section. The or each handle may be substantially U-shaped. The angle between the transverse section and each of the upright sections may be substantially a right angle. Each upright section may extend substantially perpendicularly to the plane of the frame, or each upright section may extend at an angle of less than or greater than 90 degrees to the plane of the frame.

(98) Providing a handle with multiple connection points with the frame, i.e. via the upright

sections, may increase the stability of the handle. In addition, the fastening load at each of the connection points would not be required to be as great compared to a handle comprising a single connection point. Such a handle also provides multiple grip positions with a single handle, such as an overhand or underhand horizontal grip on the transverse section, or a vertical grip on the upright sections.

(99) The exercise device may comprise straps attached to the frame. The straps may be configured to attach the device to the user. The straps may be configured to suspend the device from the body of the user. The straps may be configured to be worn over the shoulders of the user. The straps may be configured to support the exercise device. The straps may suspend the exercise device from the user's shoulders. The straps may enable the user to release the device without the device dropping to the floor. The straps may be adjustable straps. The straps may be non-adjustable straps.

Adjustable straps may enable the user to adjust the position of the exercise device relative to the user's body. The straps may be configured to support the device relative to the user's body. The straps may be connected to, or configured to be connected to, the frame. The straps may be adjustable straps configured to vary the position of the device relative to the user's body.

(100) The exercise device may be configured to provide a variable moment of inertia of the device. The exercise device may be configured to provide a variable moment of inertia about a central axis of the frame. The exercise device may be configured to provide a variable moment of inertia about a non-central axis of the frame. The user may vary the moment of inertia about a rotational axis of the device in order to vary difficulty of exercise experienced when using the device. The moment of inertia may be increased and decreased by respectively adding and subtracting weight from the device.

(101) The frame of the exercise device may be constructed using hollow tubes. The innermost frame section may be constructed from hollow tubes. The outer most frame section may be constructed from hollow tubes. All of the frame sections may be constructed from hollow tubes. A combination of frame sections may be constructed from hollow tubes. The frame may be constructed using hollow tubes which can be filled with a liquid such as water in order to increase the moment of inertia about an axis of rotation of the device.

(102) The frame of the exercise device may be configured to have weights attached to it in order to increase the moment of inertia about an axis of rotation of the device. The frame may be configured such that weights can be suspended from the innermost frame section. The frame may be configured such that weights can be suspended from the outermost frame section. The frame may be configured such that weights can be suspended from any, or a combination of, frame sections.

(103) The exercise device may comprise a resistance member attached to the frame. The resistance member may be attached to the frame via the attachment point. A first end of the resistance member may be attached to the frame. A second end of the resistance member may be configured to be attached to an external resistance point (such as a wall, a floor, a door, or a pulley system of weights). The exercise device may comprise a plurality of resistance members attached to the frame. The resistance members may be attached to the frame via a single attachment point or by respective, separate attachment points. The resistance members may be configured to be attached to a single resistance point.

(104) The resistance member may be a resistance cable, such as a rope, string, an elastic cable or in some other elongate form. The resistance member may also be a closed loop, such as in an elastic band. There may be a plurality of resistance cables for a user to choose from. The different resistance members may provide different levels of resistance. Different resistance members may have different spring constants. Some resistance members may be non-extendable, meaning he or she is resistant to extension under the tension provided by ordinary human force.

(105) The user may choose from one of the plurality of resistance members in order for the device to provide a desired torque against which the user can exercise. The resistance member may be configured to wrap around the outer perimeter of the frame multiple times to increase the tension

throughout the resistance member, which would increase the torque against which the user exercises. A plurality of resistance members may be attached to the frame at the same time. A plurality of resistance members may be attached to a plurality of different attachment points at the same time. A plurality of resistance members may be attached to the same attachment point at the same time.

(106) The frame may comprise a receiving means in the form of a channel configured to receive the resistance member attached to the frame. The receiving means may prevent the resistance member from losing contact with an outer perimeter of the frame. The receiving means may be configured to support the resistance member. The receiving means may be a receiving groove. The receiving means may be present along all or most of an outer perimeter of the frame. The receiving means may prevent the resistance member from slipping off the frame and hitting the user.

(107) The resistance point, which may be attached to the second end of the resistance member, may be a stationary resistance point. The stationary resistance point may be a wall, a floor, or an object which is sufficiently heavy such that it will not be moved by a force from the resistance member. The resistance member may comprise a loop of clip in order to be retained by the resistance point or may be simply tied and/or knotted around the resistance point (such as when the resistance member comprises a rope or band).

(108) The resistance point which may be attached to the second end of a resistance member may be a moveable resistance point. The moveable resistance point may be a pulley weight system. The resistance point may be attached to a system such that the resistance member is configured to apply a variable force to the device. The user may increase the weight on the pulley weight system in order to experience a greater torque to exercise against. The resistance point may be an exercise device already known in the art.

(109) The user may attach the second end of the resistance member to a pulley weight system, and in this situation, the user may use a non-stretchable resistance member. The device may be configured to be used in combination with pre-existing devices. The user may vary the weight system in order to increase or decrease the tension throughout the resistance member. The user may increase or decrease the tension throughout the resistance member in order to experience a different amount of torque to exercise against.

(110) The user may stand closer or further away from the resistance point in order to increase or decrease the tension throughout the resistance member. The user may choose to wrap the resistance member around the outer perimeter of the device a multiple number of times in order to increase the tension throughout the resistance member.

(111) The user may rotate in a first direction to exercise a first group of muscles, and may rotate in a second direction to exercise a second group of muscles. The user may use the device whilst in a standing position in order to exercise. The user may keep the device substantially perpendicular to his or her upper body whilst exercising. The user may rotate the device by rotating his or her upper body only. The user may rotate the device by rotating his or her arms relative to his or her body. The user may rotate his or her body forward at the hips or knees whilst using the device in order to strengthen the specific muscle groups associated with swinging a racket, club or bat. The user may use the device in various positions in order to exercise different muscle groups.

(112) A pendulum, or swinging member, may be provided for use with or as part of the exercise device. The pendulum may be provided instead of or in addition to a resistance member. The pendulum may be configured to swing when the user rotates the device using their core. The forces exerted by the pendulum on the user may further condition or strengthen core muscles, and train the user. The pendulum may oscillate in use and a user's attempt to control the oscillation, when the device is in use, may provide guidance to the user for improving his or her form (wherein form may refer to exercising in a particular way and/or in a way that is in imitation of performing a particular sporting action—such as a golf swing).

(113) The device may be configured to predominantly exercise the user's core muscles during

exercise. The device may be configured to also exercise other muscles groups during exercise. (114) The device may be configured such that if the user accidentally releases the device during exercise, it will not be repelled towards the resistance point to avoid someone being injured or something being damaged.

(115) The use of the term “diameter” throughout the description and claims should be taken to mean the widest point when referring to a non-circular feature.

(116) According to another aspect of the invention there may be a kit of parts, the kit of parts comprising an exercise device according to the first aspect and a stability training device. The stability training device may be of the type that a user stands on in use. The stability training device may be designed to be deliberately unstable when a user is standing on the stability training device, so as to train the core muscles of the user as the user attempts to maintain stability. The stability training device may comprise a board or a cylinder or curved surface for standing on, and the stability training device may be compressible, for example a compressible cellular or foam structure . . .

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Examples of the invention will now be described, with reference to the accompanying drawings, in which:

(2) FIG. 1 shows a photograph piece of exercise equipment known in the art;

(3) FIG. 2 shows a photograph of another piece of exercise equipment known in the art;

(4) FIG. 3 shows a photograph of an exercise device in use;

(5) FIG. 4 shows a simplified plan view of the exercise device of FIG. 3;

(6) FIG. 5 shows a simplified front view of the exercise device of FIG. 3;

(7) FIG. 6 shows a simplified perspective view of the exercise device of FIG. 3;

(8) FIG. 7 shows a simplified front view of the exercise device of FIG. 3 being prepared for use;

(9) FIG. 8 shows a simplified front view of the exercise device of FIG. 3 in use;

(10) FIG. 9 shows a simplified front view of another exercise device in use;

(11) FIG. 10 shows a simplified front view of another exercise device in use;

(12) FIG. 11 shows a simplified plan view of the exercise device of FIG. 9 in use;

(13) FIG. 12 shows a simplified plan view of the exercise device of FIG. 10 in use;

(14) FIG. 13 shows a simplified plan view of another exercise device;

(15) FIG. 14 shows a simplified plan view of another exercise device;

(16) FIG. 15 shows a simplified plan view of an exercise device in use by a user;

(17) FIG. 16 shows a cross section of a part of an exercise device;

(18) FIG. 17 shows a cross section of a part of another exercise device;

(19) FIG. 18 shows a cross section of a part of another exercise device;

(20) FIG. 19 shows a cross section of a part of another exercise device;

(21) FIG. 20 shows a cross section of a part of another exercise device;

(22) FIG. 21 shows a cross section of a part of another exercise device;

(23) FIG. 22 shows a simplified plan view of an exercise device being prepared for use by a user;

(24) FIG. 23 shows a simplified plan view of a user using the exercise in a first position;

(25) FIG. 24 shows a simplified plan view of a user using the exercise in a second position;

(26) FIG. 25 shows a simplified plan view of a user using the exercise in a third position;

(27) FIG. 26 shows a simplified plan view of a user using the exercise in a fourth position;

(28) FIG. 27 shows a force diagram representing the forces applied when an exercise device is in use;

(29) FIG. 28 shows a simplified schematic of an exercise device in use in a first arrangement;

(30) FIG. 29 shows a simplified schematic of an exercise device in use in a second arrangement;
(31) FIG. 30 shows a simplified schematic of an exercise device in use in a third arrangement;
(32) FIG. 31 shows a simplified schematic of an exercise device in use in a fourth arrangement;
(33) FIG. 32 shows a simplified schematic of an exercise device in use in a fifth arrangement;
(34) FIG. 33 shows a simplified schematic of an exercise device in use in a sixth arrangement;
(35) FIG. 34 shows a simplified schematic of an exercise device in use in a seventh arrangement;
(36) FIG. 35 shows a simplified schematic of an exercise device in use in an eighth arrangement;
(37) FIG. 36 shows a simplified schematic of another exercise device;
(38) FIG. 37 shows a perspective view of another exercise device;
(39) FIG. 38 shows perspective view from the front of the exercise device of FIG. 37;
(40) FIG. 39 shows another perspective view from of the exercise device of FIG. 37 from an angle showing the underneath of the device;
(41) FIG. 40 shows a perspective view of the exercise device of FIG. 37, highlighting a section;
(42) FIG. 41 shows a cross section through the highlighted section of FIG. 40;
(43) FIG. 42 shows a perspective view of the exercise device of FIG. 37, highlighting another section;
(44) FIG. 43 shows a cross section through the highlighted section of FIG. 42;
(45) FIG. 44 shows a magnified view of the highlighted section of FIG. 42;
(46) FIG. 45 shows a partially-exploded side view of the exercise device of FIG. 37;
(47) FIG. 46 shows a cross section of a portion of the exercise device of FIG. 37;
(48) FIG. 47 shows a perspective view of the exercise device of FIG. 37, highlighting another section;
(49) FIG. 48 shows a magnified view of the highlighted section of FIG. 47 with some parts removed for clarity;
(50) FIG. 49 shows a magnified view of the highlighted section of FIG. 47 with some other parts removed for clarity;
(51) FIG. 50 shows a perspective view of the exercise device in use in a first configuration;
(52) FIG. 51 shows a perspective view of the exercise device in use in a second configuration; and
(53) FIG. 52 shows a perspective view of the exercise device in use in a third configuration;
(54) FIG. 53 shows a simplified schematic view of the exercise device in use in another arrangement;
(55) FIG. 54 shows a simplified schematic view of the exercise device in isolation when in use in the arrangement of FIG. 53;
(56) FIG. 55 shows screenshots from a video of the device when used in the arrangement of FIG. 53;
(57) FIG. 56 shows screenshots from a video of the device when used in another arrangement;
(58) FIG. 57 shows a sectional view through a part of the device;
(59) FIG. 58 shows an attachment point of the device;
(60) FIG. 59 shows another attachment point of the device;
(61) FIG. 60 shows an attachment point of the device with a carabiner attached;
(62) FIG. 61 shows a view from underneath the device;
(63) FIG. 62 shows a schematic front view of an exercise device;
(64) FIG. 63 shows a schematic plan view of an exercise device;
(65) FIG. 64 shows a schematic plan view of an exercise device;
(66) FIG. 65 shows a front view of an exercise device;
(67) FIG. 66 shows a close-up view of a portion of the device of FIG. 65;
(68) FIG. 67 shows a further close-up view of a portion of the device of FIG. 65;
(69) FIG. 68 shows a further close-up view of a portion of the device of FIG. 65;
(70) FIG. 69 shows a rear view of the device of FIG. 65;
(71) FIG. 70 shows a plan view of the device of FIG. 65;

- (72) FIG. **71** shows a schematic plan view of the exercise device of FIG. **65**;
- (73) FIG. **72** shows a schematic front view of a portion of the exercise device of FIG. **65**;
- (74) FIG. **73** shows the same view as FIG. **71** with the exercise device in a different position;
- (75) FIG. **74** shows a schematic front view of a portion of the exercise device of FIG. **65** in the configuration shown in FIG. **73**;
- (76) FIG. **75** shows the same view as FIG. **71** with the exercise device in a different position;
- (77) FIG. **76** shows a schematic front view of a portion of the exercise device of FIG. **65** in the configuration shown in FIG. **75**;
- (78) FIG. **77** shows a schematic plan view of another embodiment of the device of FIG. **65**;
- (79) FIG. **78** shows a schematic plan view of a portion of the device of FIG. **77**;
- (80) FIG. **79** shows a schematic side view of the pulley of the device of FIG. **77**;
- (81) FIG. **80** shows a schematic side view of a portion of the device of FIG. **77** comprising the pulley; and
- (82) FIG. **81** shows the device of FIG. **65** in use.

DETAILED DESCRIPTION

(83) FIG. **1** shows a first exercise machine **10** that is currently available and known in the prior art. The exercise machine is a weight and pulley based machine and comprises handles **11**, cables (not visible in FIG. **1** as encased within the machine), and weights **12**. The cables are attached to the handles **11** at a first end and he or she is attached to the weights **12** at a second end. Various exercises can be performed using the apparatus but fundamentally each exercise consists of pulling the handles **1** to exercise against the weight of the weights **12**. In one exercise a user holds the handles **1** and pulls them across his or her body. As the handles **11** are pulled, the weights **12** are lifted via the cables. The weights **12** provide a resistance force through the cable **13** to the handles **11** which the user resists against as a means of exercise. The user can vary the amount of weight attached to the second of the cables **13** in order to vary the difficulty of the exercise. The user can also vary the position of the handles **11** relative to the user's body by adjusting pivotable arms **13** to which the handles **11** are attached.

(84) FIG. **2** shows a second exercise machine **20** that is currently available and known in the prior art, which is also a weight and pulley based machine. The second exercise machine **20** operates in operable in a similar manner to the first exercise machine **10**, and also weights **22** and cables **23**. The cables **23** are attached to the weights **22** (which are configurable for different resistances) at one end and to an attachment point **21** at the other end. The attachment point **21** may receive a handle of various types in dependence on a selected exercise type. In this machine **20** the user can also vary the position of the attachment points **21** (and therefore the handles) relative to the user's body through vertically raising or lowering the attachment points in order to accommodate for a range of different exercises, and to accommodate for a range of users with different heights.

(85) In each of the prior art exercise machines **10**, **20** the user is provided with control over the positioning of his or her hands and arms relative to his or her body. Whilst this freedom can provide for a variety of exercises to be performed it can also lead to injury or ineffective exercises due to poor technique and body positioning.

(86) An example of an exercise device **100** for improving core strength is shown being used in the photograph of FIG. **3**. The device **100** can be used in combination with the exercise machines **10**, **20** of the prior art or can be used as a standalone piece of exercise equipment.

(87) A simplified, plan view of the exercise device is shown in FIG. **4**. The exercise device **100** comprises a frame **110** which defines a central aperture **112**. A user may stand within the central aperture **112** in order to use the exercise device **100** as shown in FIG. **3**.

(88) The frame **110** comprises a first frame section **116** and a second frame section **118**. The second frame section **118** has a greater outer diameter than the first frame section **116**. An inner diameter of the first frame section **116** defines an inner diameter of the frame **110**. The outer diameter of the second frame section **118** defines an outer diameter of the frame **110**. The first frame section **116**

and the second frame section **118** share a common central axis. In this example, in which the first and second frame sections are circular, the first frame section **116** and the second frame section **118** are therefore concentric. In other examples, the frame may comprise only a single frame section or any multiple number of frame sections. In other examples, the frame sections **116**, **118** may not share a common centre.

(89) The frame **110** is substantially circular. The first frame section **116** and the second frame section **118** are also substantially circular. In other examples the frame **110** may have another shape, such as any regular polygon or any irregular polygon, a conic section or a pear shape. Different shaped frames **110** may be used in order for the user to exercise different muscles or muscle groups with the device.

(90) The inner diameter of the first frame section **116** is configured to be greater than the girth of a user, such that the user can fit his or her torso within the central aperture **112** of the frame **110**. The outer diameter of the second frame section **118** is not too large so that the user can reach the second frame section **118** when he or she is positioned within the central aperture **112**.

(91) The first frame section **116** and the second frame section **118** are attached together via a plurality of fasteners **120**. FIG. 1 shows an example comprising eight fasteners **120**, but any number of fasteners may be used instead. Preferably at least three fasteners are used. Using at least three fasteners substantially equally spaced, may provide stability of the first section relative to the second section. In the present example, the fasteners **120** are adjustable fasteners. The fasteners are adjustable through the use of a ratchet mechanism. Other adjustment systems may also be used, such as adjustable straps, turnbuckles, or any other suitable means.

(92) In other examples, the frame sections **116**, **118** may be connected via non-adjustable fasteners, or a combination of adjustable and non-adjustable fasteners. The first frame section **116** is adjustable in this example, and its diameter can be varied by adjusting the adjustable fasteners **120**. The diameter of the first frame section **116** may be adjusted in order to fit proximally around the differently sized waists of different users.

(93) The first frame **116** section may comprise an elastic ring. The elastic ring may be constructed such that it has a diameter which is large enough to fit proximally around the torso of the user when there are no forces being exerted on it. The elastic ring **116** may be configured such that its diameter increases when there are forces being exerted on it via the fasteners **120**. The fasteners **120** may be adjustable fasteners, and may be adjusted in order to adjust the force exerted on, and the inner diameter of, the elastic ring **116**. The elastic ring **116** may be configured such that its diameter can be increased and/or decreased to a size at which it can fit proximally around the torso of users with various girths.

(94) The frame **110** and its components (**116**, **118**, **120**) can be made from any suitable material such as plastic, wood, or metal. The materials used are sufficiently strong so that he or she does not significantly bend or deform under his or her own weight or when in use. The materials are lightweight so that a user is able to lift the device **100** without great effort. The frame **110** may be constructed using hollow tubes in order to reduce its weight. In this example the frame **100** and its components are formed from a polymer. Each part is formed from polyethylene. More specifically the outer frame section **118** is formed from high-density polyethylene and the other components are formed from low-density polyethylene. Different materials may be used in dependence on design criteria and the envisaged end user. For example, a device designed for an adult to use may be stronger (and potentially heavier) than a device designed for use by a child. Lightweight materials are preferred, having sufficient structural rigidity for the frame sections (plastics and metal, e.g. aluminium, for example).

(95) FIG. 5 shows a simplified front view of the exercise device of FIG. 4. The second frame section **118** of the exercise device comprises a top section **122** and a bottom section **124**. The top section is connected to the bottom section via connecting rods **126**. The connecting rods in this example form a channel **130**. The exercise device comprises handles **114** attached to the frame **110**.

In this example a pair of handles is provided. The pair of handles provides one handle for each hand of the user. In other examples a single handle may be provided or more than two handles may be provided. A single handle may allow a single point of contact for the user, or for the user to place both hands in a central location. Multiple handles (e.g. 3, 4, 5, 6 or multiple pairs of handles) may provide a means for a user to vary his or her arm positioning, which may allow them to vary the exercise types and which muscle groups are exercised when using the device. The handles provide a means by which the user can support the exercise device **100** in position about the torso of said user. The handles **114** are connected to the second frame section **118**, but in other examples, they may be connected to the first frame section **116** or any other frame section. Handles may be provided on both its first and second frame sections.

(96) The handles **114** provide a grip for the user. The handles **114** may be profiled or coated with an appropriate material, such as rubber, in order to provide comfort for the user when grasping the handles **114**. In the present example, the handles **114** are shown as protrusions extending from the second frame section **118**. However, in other examples, the handles may be integrally formed as part of the frame **110**. The handles **114** may be formed from indents in the frame sections (**116**, **118**) which are configured to receive the user's fingers. The handles **114** may be constructed from, or coated with, a material which increases the grip a user can get on the handles **114**.

(97) FIG. **6** shows a simplified, perspective view of the exercise device **100** of FIGS. **4** and **5** in which some features have been omitted for clarity.

(98) FIG. **7** shows a simplified front view of the exercise device of FIGS. **4-6** being prepared for use. There is a resistance member **142** attached at a first end of the resistance member **142** to one of the connecting rods **126** of the device. In this example, the end of the resistance member **142** is tied around the connecting rod **126**, the connecting rod forming a connection point of the device.

(99) FIG. **8** shows a simplified front view of the exercise device of FIGS. **4-6** in use. The resistance member **142** is wrapped around majority portion of the channel **130**. The resistance member **142** abuts the connecting rods **126** in this example. The connecting rods **126** forming the channel which is configured such that the resistance member will not slip out of contact with the outer perimeter of the exercise device **100** when the resistance member is under tension. The channel is not always formed from connecting rods **126**, as shown in the example in FIG. **9**.

(100) The resistance member may be supplied separately or with the frame **110** of the exercise device **100** and be attachable to the frame. The resistance member **142** may be formed integrally with the frame **110** in other examples. A first end of the resistance member **142** is attached to the frame **110** via an attachment point **140**. In other examples, there may be multiple attachment points. In this example the attachment point **140** is positioned on the second frame section **118**. In other examples, the attachment point(s) **140** may be positioned on the first frame section. Having multiple attachment points allows for the position of the resistance member relative to the user to be modified. This can allow for further configuration in order to account for different exercise regimes. The attachment point (or attachment points) may also be re-positionable on the frame as an alternate or supplementary means to provide further configuration options. The attachment point may be re-positionable using a clamping means, by which the attachment point clamps to the frame, or by any other suitable, releasable attachment means. Various attachment points are envisaged. The attachment point used is configured to be co-operable with the resistance member and/or vice-versa. For example, a hook-and-eye system may be used, in which the resistance member comprises a loop (i.e. the eye) at a first end and the attachment comprises a hook, peg, or similar, over which the loop may be secured. The attachment point may comprise a clamp, configured to be releasably clamped onto the first end of the resistance member,

(101) In this example, the resistance member **142** is a stretchable member, such as an elastic resistance band. The resistance force to the user twisting his or her torso is provided through stretching the resistance member. The strength required of the user to exercise with the member being proportional to the spring constant of the resistance member **142**. In other examples, the

resistance member **142** may be a relatively inelastic resistance cable (i.e. one that is not meant to be stretched during use), for example a steel or nylon cable such as steel wire rope often used in fixed gym equipment. In these examples resistance is generally not provided by the resistance member, but through a system (e.g. a weight or separate spring) to which the resistance member may be attached at its second end.

(102) The second end of the resistance member **142** is configured to be attached to an external resistance point. The external resistance point may be stationary, fixed point such as a wall, a floor, a post, or a door handle (on a closed door) or may be a dynamic resistance point such as a pulley weight system. For an exercise device using a stretchable member a fixed resistance point is used. For an exercise using an in-elastic resistance member a dynamic resistance point is used.

(103) The resistance member **142** may be wrapped or partly wrapped around the perimeter of the frame **110** when the device is in use or when in storage. The resistance member may be wrapped around the second frame section **118** once or any number of times. Wrapping the resistance member around the device can be used to increase the tension throughout the resistance member. Wrapping the resistance member around the device also provides a storage means, by which the resistance member can be kept with the frame **110** for improved storage and transportation (compared with keeping the resistance member **142** separate from the frame **110**). Wrapping the resistance member around the perimeter of the frame may apply an inwardly directed force (i.e. towards the central axis of the device) which may also assist the resistance member with locating on the device and reducing the likelihood of the resistance member becoming unseated from the device during use.

(104) The exercise device **100** comprises a receiver which is configured to receive the resistance member **142** when the resistance member is wrapped around the frame **110**. The receiver is comprised within the frame **110** in this example. The second frame section **118** comprises connecting rods (not shown in FIG. 2) that form the receiver in which the resistance member **142** is held. The connecting rods connect an upper section of the frame to a lower section of the frame. When the resistance member **142** is wrapped around the frame **110**, the resistance member **142** may apply a rotational force equally along the entire perimeter of the frame **110**. In other examples the resistance member **142** may be partially wrapped around the perimeter of the frame **110**, the resistance member **142** may apply a rotational force continuously along the section of the frame **110** which is in contact with the resistance member **142**.

(105) FIG. 9 shows a simplified front view of another exercise device **200** in use. In this example, the frame **210** comprises a top section **222**, a bottom section **224**, and a joining section **210**. The joining section **210** defines a central aperture (not shown). The joining section **210** has a curved outer perimeter which defines receiver. The device **200** comprises a connection point **226** which provides a means by which a user can attach a resistance member **142** to the device **200**. The top section **222**, bottom section **224** and joining section **210** can be formed from a single piece. That is the frame can be produced as a mould, such as via injection moulding or similar.

(106) FIG. 10 shows a simplified front view of another exercise device **200** in use. In this example, the frame **210** comprises a top section **222**, a bottom section **224**, and a joining section **210**. The joining section defines a central aperture **212**. The device also contains connecting rods **216** which extend between the top section **222** and bottom section **224** of the frame **210**. The connecting rods **216** are curved in his or her centre towards the central aperture in order to define a deeper channel than the channel provided by the rods in the first example exercise device.

(107) FIG. 11 shows a simplified plan view of the exercise device of FIG. 9. The resistance member is shown in dashed line form where it is hidden by the top section **222**. The resistance member is contained within the channel and extends tangentially away from the device to a resistance point.

(108) FIG. 12 shows a simplified plan view equivalent of FIG. 10. In this example, the resistance member **142** is connected to a connecting rod **126** at a first end and is connected to an external

resistance point (not shown) at a second end. The resistance member **142** is held taught so that the resistance member **142** rests against the channel which is defined by the connecting rods **126**.

(109) FIGS. **13** and **14** show simplified plan views of other exercise devices **100**. In these examples, the central aperture **112** is not concentric with the centre of the exercise device **100**. In these examples, the lengths of the adjustable straps **120** have been increased or decreased in order to move the first frame section **116** relative to the second frame section **118**.

(110) FIG. **15** shows a simplified plan view of an exercise device **100** being used by a user **150**. The user **150** is supporting the device **100** by resting the frame **110** against his or her forearms. The user **150** is holding the handles in order to exert an opposing torque on the device **100**. In this example, the second end of the resistance member **142** is connected to an external resistance point **144** which is a stationary post.

(111) FIGS. **16-21** shows the examples of forms of connecting rods **126** in greater detail than in earlier figures. The figures provide a cross-sectional view of the connecting rods **126**. The connecting rods **126** are connected between the top section **122** and the bottom section **124** of the second frame section **118**. The connecting rods **126** are configured to allow a resistance **142** member to rest against them.

(112) The top section **122** and the bottom section **124** of the second frame section **118** are connected via the connecting rods **126**. The connecting rods **126** can be curved, such that the centre of the connecting rods **126** (the centre being the point midway between the top section **122** and the bottom section **124**) is closer to the centre of the exercise device **100**.

(113) The connecting rods may be constructed from any suitable material which will not break under the inwardly (i.e. towards the central axis of the frame) directed force exerted by the resistance member **142**. In the example shown, the top section **122** and bottom section **124** of the second frame section are hollow. In other examples the top section **122** may be hollow and the bottom section **124** may not be hollow, or vice versa. In other examples, the top and bottom sections **122**, **124** may both be solid throughout.

(114) The curvature of the connecting rods **126** creates a channel which provides a receiving means in which the resistance member **142** is received. In this example, the channel **130** is formed by the groove. In other examples, the receiving means **130** may be a receiving tube or other receiver. The channel **130** prevents the weight of the resistance member **142** from causing the resistance member **142** to fall away from the second frame section **118**.

(115) The connecting rods **126** may also be used as an attachment point to attach the resistance member **142** to the device **100**. In such examples, the first end of the resistance member **142** can be tied around one of the connecting rods **126** in order to connect the resistance member **142** to the exercise device **100**.

(116) In other examples the channel **130** may be constructed from a single section. The single section may be of semi-circular cross section wherein the centre of the cross section curves towards the centre of the frame **110**. The semi-circular cross section may be configured to support the resistance member **142**. In other examples, the single section may be of circular cross section (i.e. a tube) configured to contain a portion of the resistance member **142**.

(117) The first frame section **116** (shown in FIGS. **1** and **2**) may be constructed similarly to the second frame section **118** shown in FIG. **3**.

(118) FIG. **22** shows an example of a user **150** in a position ready to use the exercise device **100**. The user **150** is positioned within the central aperture **112** of the frame **110**. The frame **110** of the exercise device **100** is being held by the user **150** via the handles **114** which are attached to the first frame section. The first frame section is not shown in FIG. **22** for clarity. The resistance member **142** is provided between the exercise device **100** and a resistance point **144**. The frame **110** is being held by the user **150** at a level height with the torso of the user **150**.

(119) FIGS. **22-26** shows a user **150** using the device **100** to exercise. The resistance member **142** used in this example is an elastic cable. A first end of the resistance member **142** is attached to the

second frame section **118** via an attachment point **140**. A second end of the resistance member **142** is attached to a resistance point **144** which is external from the exercise device **100**. In the example shown, the resistance point **144** is a stationary post, which is an example of a fixed resistance point. (120) The user **150** in FIG. **22** is facing directly towards the resistance point **144**. The user **150** is stood at a distance from the resistance point **144** such that the resistance member **142** is extended beyond its natural length and so there is tension throughout the resistance member **142** which provides an elastic resistance force, but not enough tension to exert a force that can forcibly move the user **150**. The user **150** may stand closer or further away from resistance point **144** in order to increase or decrease the tension throughout the resistance member **142**.

(121) The user may use different resistance members **142** to increase or decrease the tension throughout the resistance member **142**. The user may use a resistance member **142** with a shorter natural length in order to increase the tension throughout the resistance member **142** without having the stand in a different position. In contrast, the user **150** may use a resistance member **142** with a greater natural length in order to decrease the tension throughout the resistance member **142** without having to stand in a different position.

(122) Further, the user **150** may use a resistance member **142** with a greater spring constant in order to increase the tension throughout the resistance member **142**. In contrast the user **150** may use a resistance member **142** with a smaller spring constant in order to decrease the tension throughout the resistance member **142**. The user may choose to increase or decrease the tension throughout the resistance member **142** in this way in order to vary the intensity of exercise he or she experiences.

(123) In FIG. **23** the user **150** has turned around through an angle of 180° anti-clockwise from the starting position in FIG. **22** and has rotated the device **100** simultaneously. The user **150** grips the handles (not shown) whilst turning so that the device turns through the same angle as the user. The tension in the resistance cable **142** increases upon rotation because it extends further beyond its natural length. The tension in the resistance cable **142** produces a resistance force which acts on the second frame section **118** via the connection point (not shown), and it acts in a direction towards the fixed resistance point **144**.

(124) The component of the resistance force which acts tangentially to the perimeter of the second frame section **118** creates a clockwise torque, indicated by the arrow **152**. The user **150** resists against this torque and uses it as a means of exercise. The user **150** may twist in an anti-clockwise direction to increase the amount of torque being applied or reduce the torque through twisting clockwise. As the user twists through a greater angle, the torque applied to the device **100** increases. In this example the user twists between the positions shown in FIGS. **23** and **24** in order to exercise.

(125) In order to exercise the other side of his or her body the user can turn the other way from the starting position shown in FIG. **22**. As shown in FIGS. **25** and **25** an anti-clockwise torque **152** is provided against which the user can exercise through twisting in a clockwise direction.

(126) By twisting it is meant that the user keeps his or her feet in position or at least does not move his or her position much, and the movement comes from turning of the torso. Such actions are common in many sports, particularly sports which involve striking—such as golf, hockey and racquet sports.

(127) The user **150** may twist in clockwise and anti-clockwise directions periodically as a means of exercise. The user **150** may rotate through small angles to repeatedly exercise a specific muscle group. The user may rotate through larger angles to exercise a wider variety of muscle groups.

(128) In other examples, the user **150** may rotate clockwise from the starting position. In such examples, the torque which the user **150** exercises against would then be in an anti-clockwise direction.

(129) In some situations the user may turn the device through an angle greater than 360° in order to get into a starting position. In such a situation, the resistance member **142** will wrap around the perimeter of the second frame section **118**. As the user **150** rotates, the resistance member **142** may

be supported within the channel **130**. The channel may decrease the chance of the resistance member **142** losing contact with the second frame section **118** than if the resistance member simply wraps around a perimeter of the device.

(130) In the examples shown in FIGS. **23-26**, the user is using the centre of the device **100** as the axis of rotation. In other examples, the centre of rotation may be displaced from the centre of the device.

(131) In other examples, the resistance member **142** may be non-extendable and the resistance point **144** may be a pulley weight system. Examples are shown in FIGS. **28-29**. In these examples, the resistance force does not vary with the angle through which the user **150** rotates. Instead, in this example, the resistance force which is applied to the second frame section **118**, and hence also the torque, is dependent on the amount of weight being lifted by the pulley weight system **144**.

(132) As the user **150** rotates the device **100**, the attachment point **140** moves the resistance member **142** via the first end of the resistance member **142**. The second end of the resistance member **142** is attached to the pulley weight system **144**. As the user rotates, the resistance member **142** exerts a force on the pulley weight system **144** in order to lift a set of weights. The user may periodically rotate clockwise and anti-clockwise in order to repeatedly lift the weights as a means of exercise.

(133) The non-extendable resistance member **142** is made from a material which is sufficiently strong to avoid it snapping when a large force is imposed on it from the pulley weight system.

(134) FIG. **27** shows how the moment about the centre of rotation **102** of the device **100** can vary according to which frame section **116**, **118** the force is applied to. The first frame section **116** is a distance d_1 away from the centre of the device **100**. The second frame section is a distance d_2 away from the centre of the device **100**.

(135) When a force is applied to the first frame section **116** as indicated by arrow A, the moment produced about the centre of the rotation **102** of the device **100** is given by the equation:

$\text{Moment} = d_1 \times \text{Size of force A}$. Similarly, When a force is applied to the second frame section **118** as indicated by arrow B, the moment produced about the centre of the rotation **102** of the device **100** is given by the equation: $\text{Moment} = d_2 \times \text{Size of force B}$. As such, the same moment can be applied by applying less force when the force is applied to the second frame section **118** rather than the first frame section **116**. Similarly, the same moment can be applied by applying a greater force when the force is applied to the first frame section **116** rather than the second frame section **118**. A user may use handles and/or attachment points positioned on the various frame sections according to this principle.

(136) FIGS. **28** and **29** show a user using the device with a weighted pulley exercise machine (which may be an exercise machine **10**, **20** such as those shown in FIGS. **1** and **2**. In FIG. **28** the device has a concentric aperture to the device. In FIG. **29** the aperture is off-centre. Differing the axes of the aperture and the frame can allow for different exercise types to be performed and also to provide different weight profiles. For example a circular device with a concentric aperture may provide a more uniform resistance, whereas a non-concentric aperture may provide a non-uniform resistance profile. A non-uniform resistance profile can be useful for simulating sporting movements (e.g. at different points of a swing of a racquet, club or bat)

(137) FIG. **30** shows a simplified schematic of an exercise device in use with a fixed resistance point. In this example, there are straps **128** connected to the second frame section **118**. The straps are being worn over the shoulders of the user **150** in order to support the device **100**. In other examples, the straps **128** may be connected to any other frame section, such as the first frame section (not shown). The straps **128** are constructed from a durable fabric. Suitable materials include polyester, leather or cotton or any other suitable material. Padding may be provided on the straps in a region that is configured to sit on the user's shoulders in use in order to provide improved comfort over an unpadded strap. The straps **128** are constructed of a material that is lightweight and that is comfortable for a user to wear. The straps **128** are constructed to have

enough tensile strength such that the straps do not distort or snap when a force is exerted on them through the weight of the device **100**. The shoulder straps may be adjustable. Adjusting the shoulder straps may allow the device to be configured to fit different torso lengths. The adjustability may also provide the option to position the device to exercise different muscle groups. (138) FIG. **31** shows a simplified schematic of an exercise device **100** in use with a fixed resistance point. In this example, the central aperture **112** which receives the user is not concentric with the centre of the device **100**.

(139) FIGS. **32** and **33** show simplified schematics of an alternative way of a user using the exercise devices in FIGS. **30** and **31** respectively. In these examples, the user **150** has rotated his or her torso forward through a small angle in order to exercise different muscles. The user can therefore alter his or her body position to mimic the positions used in various sports.

(140) FIGS. **34** and **35** show simplified schematics of a user using the exercise device in alternative ways. In these examples, the user **150** has rotated the device **100** so that it is no longer parallel with the length of the user's **150** body and instead sits around the shoulders. This can allow a user to exercise different muscles groups, such as the trapezoidal muscles.

(141) In FIG. **35**, there are handles **114** above attached to the top section **122** and below attached to the bottom section **124** of the device **100**. This enables a user **150** to exercise different muscles and muscle groups. This may allow a user to replicate other sports which also require core strength but also differing movement profiles. In this case such a sport may be canoeing or kayaking.

(142) In some examples the device is not used with an external resistance point in order to provide a resistance torque. Instead, a user can simply exercise against a moment of inertia of the device. In order to vary the moment of inertia the resistance member can be provided to increase mass rather than to be fixed to an external resistance point. Such an example is shown in FIG. **36** shows a simplified schematic of an alternative exercise device **100**. In this example, the resistance member **142** is a rope comprising weights **143**. The weighted rope **142** is wrapped around the outer perimeter of the device **100** to increase the moment of inertia about the centre of the device.

(143) Another example of an exercise device **500** is shown in FIGS. **37-39**. In this example the exercise device **500** has a 'walk-in' design rather than an 'over-the-head' design as in the previous examples (although a user could still use this device by passing it over his/her head when the device is fully assembled). The frame **510** of the device **500** does not fully encompass the user when used, but rather has a crescent form into which a user can step. Other forms, such as semi-circular or arc may also be used in place of the crescent shape. The frame is of lightweight construction and has a plurality of structural webs **520** that provide structural rigidity to the frame. The frame **510** is constructed from two plastic moulded components split horizontally. Forming from the two components allows for easier moulding and forming of the features of the device than if the frame **510** was formed from a single component. A support strap **590** is attached to the frame to close the aperture **512** in which the user places themselves. The support strap **590** comprises a cushioned section **591**. The cushioned section **591** provides lumbar support when in use. The cushioned section **591** is slidable along the support strap **590** to provide easy adjustability to position the support in the appropriate position. As in the previous examples this device **500** of this example comprises an attachment point **526** and a channel **530** in which a resistance device **142** can be attached and located within respectively.

(144) The channel **530** is extended in at least one area to offer more protection to the user in case of accidental disconnection of the resistance member **142**. The extension is provided through the addition of at least one releasably attachable shield **540**. The shield is substantially concave to locate the resistance member in the same manner as the channel **530**. The shield **540** extends vertically in both directions to provide protection in either direction. There are preferably a plurality of shields, at least one for each of the left and right sides of the device **500**. Each shield **540** clips into a hole **570** formed in the frame **510**. The holes **510** also provide light weighting to the device and walls of the holes provide stiffness to the device **500**. Adjustable handles **514** are

provided for the user to grip when the device **500** is in use.

(145) The frame **510** has a pad **550** attached to it. The pad **510** sits opposite to the cushioned section **591** of the support strap **590** when in use. The pad **550** is a thoracic pad in that it is a pad that engages with the torso of the user when in use. In use, a user may position the device **500** such that pad **550** is adjacent to the front of the user's torso. The user may then close the aperture **512** by means of the support strap **590**, such that the cushioned section **591** of the support strap **590** is adjacent to the rear of the user's torso. Alternatively, the device **500** may be operable with the pad **550** adjacent to the rear of a user's torso. The aperture **512** does not necessarily have to be closed by means of the support strap **590** in use—the user may just put the device around the front of their torso, possibly with the pad between their torso and the frame of the device, with nothing behind their back. This may be quicker and easier to do than to pass straps behind them. In some embodiments, the support strap **590** may not be present.

(146) In addition or alternatively to the frame pad **550** and the cushioned section **591** of the support strap **590**, the frame **510** may comprise further padding. For example, padding may be provided on inner surfaces of the frame **510** which could come into contact with a user if the device **500** in use, such as at the edges of the inner section of the frame **510**.

(147) Mode detail regarding the pad **550** is shown in FIGS. **40** and **41**. FIG. **40** shows a view from the front of the device. Attachment means **560** are shown within the circled areas of the device **500**. The pad **550** is secured to the frame **510** in two locations in order to provide a secure fitting and anti-rotation. The construction of the pad is shown in FIG. **41**. The pad comprises a structural base layer **551**. The structural base layer **551** provides rigidity and stiffness to the pad. Some flexibility is preferred in construction in order to provide some flexibility in response to user movements and therefore improve comfort. This can also reduce the risk of snapping than if less flexible materials are used. Plywood is an example of a suitable material for the structural base layer **551**. A cushion layer **552** is applied to the structural base layer **551**. The cushion layer **552** provides padding and comfort to the user when the device **500** is in use. Foam padding is a suitable material for the cushion layer **552**. A top layer is provided as a cover **553**. The cover **553** encases the cushion layer **552**. The cover protects the cushion layer from ingress of sweat or other contaminants which may degrade the life of the cushion layer. The cover also provides a comfortable surface for the user to interface with. A suitable material for the cover **553** is a faux leather such as the type often used in conventional gym equipment seats. The pad **550** is attached to the frame using attachment means **560**. The attachment means **560** comprises a nut **562** and a bolt **561**. The nut is preferably a T-nut as these provide additional security and anti-rotation when used in materials such as the plywood given as an example for the structural base layer **551**.

(148) The handles **514** and their adjustment means are shown in more detail in FIGS. **42** to **44**. The handle **514** is attached to the frame **510** via a nut **517** and a bolt **516**. The bolt **516** is a threaded section that is formed or moulded within the handle **514**. The bolt **516** passes through a slot **518** in the frame **510**. The handle **514** is assembled to the frame **510** during construction. The slot is formed in a lower section **524** of the frame **510**. Once the bolt **516** is passed through the slot **518** and the nut **517** tightened on top a top section **522** of the frame **510** is fitted to the lower section **524**. This encloses the nut and prevents further adjustment. In order to provide the adjustable position of the handle a collar **515** is therefore provided with the handle **514**. The collar **515** has an internal threaded bore that engages with the thread of the bolt. Tightening or loosening the collar to close a gap **519** between the collar **515** and the frame **510** provides a releasable securing means for the handle **514**. Loosening the collar **515** allows for the handle **514** to be moved to different positions within the slot **518**. Once a suitable position is identified the collar **515** is tightened, locking the handle **514** into the preferred position. In other words to adjust the handle location, the collar is loosened at the top of the handle to release the tension between the nut, the plastic moulding and the collar. It is then possible to slide the handle to a desired position then re-fasten the collar **515** to secure the handle **514**. The collar **515** is knurled to provide grip to a user. In some

examples the handles are pivotable. A joint or hinge may be provided on the handle such that a user can vary their grip position.

(149) In some embodiments, the handle **514** may comprise a longitudinal threaded bore to receive the bolt **516**. In this way the handle **514** may be removed from the frame **510** by tightening the collar **515** to close the gap **519**, thereby gripping the collar **515** against the frame **510** and holding the bolt **516** in a stationary position, before rotating the handle **514** about its longitudinal axis so as to loosen the handle **514** from the bolt **516**. The handle **514** can be reattached by receiving the bolt **516** in the longitudinal threaded bore of the handle **514** and rotating the handle **514** about its longitudinal axis so as to tighten the handle against the bolt **516**.

(150) Removability of the handle **514** allows interchangeable handles to be used with the frame **510**. For example, a plurality of different handles may be provided. Each handle may comprise a different grip or different dimensions. The handles can be interchanged by removing a currently attached handle from the frame **510** before attaching a different handle. The interchangeable handles enable the exercise device to be used by different users with different preferences in terms of grip and handle dimensions, and/or for training for different sports with different requirements.

(151) FIGS. **45** and **46** provide more detail with regards to the fitting of the shield **540**. The shield comprises a snap hook **541**. The snap hook **541** is elastically deformable. To fit the shield **540** the snap hook **541** is inserted through one of the holes **570** of the frame **510**. The holes are formed to match the snap hook **541**. The holes **570** extend substantially radially. The snap hook **541** retains the shield **540** once inserted. To release the shield **540**, to change its position within the channel **530** for example, a user pinches the snap hook **541** together and pushes it back through the hole **570**.

(152) The attachment points for the resistance member **142** and the support strap are shown in FIGS. **47** to **49**.

(153) The attachment point **526** for the resistance member **142** is shown in FIG. **47**. In this example the resistance member **142** comprises a carabiner **528** which clips through the hole **526**. A resistance member **142** in the form of a resistance band can be passed through the hole and tied on or similar. In some examples the attachment point **526** may be positioned elsewhere on the frame **510**. The attachment point **526** can be positioned on a front facing part of the frame, rather than at the end points of the bow.

(154) The attachment point **580** for the support strap **590** is shown in FIG. **49**. The support strap **590** is adjustable to change its length. The adjustable support strap **590** provides adjustability for the device **590** for use with users of a variety of body sizes and age groups. A standard buckle fastener is shown in FIG. **49**, adjustability could also be achieved in other examples through other means, such as hook and loop (such as the type produced by Velcro (RTM)) connections for example.

(155) FIGS. **50** to **52** show various configurations of the device **500** when in-use, with the resistance member **142** attached to an external resistance point **144**. In FIG. **50** a first end of a resistance member is attached to an attachment point **526** at one end of the frame. A second end of the resistance member **142** is attached to a resistance point **144**. As the user twists in an anti-clockwise direction (when viewed from above) the resistance member sits in the channel **130** and the resistance force is provided tangentially to the device **500**. If the first end of the resistance member is attached to the other end of the bow at another attachment point then the user can twist in a clockwise direction (when viewed from above) and the resistance member sits in the channel **130** and the resistance force is provided tangentially to the device **500**. A pair of resistance members **142** can be used so that the user can rotate in both clockwise and anti-clockwise directions without having to re-position the resistance member.

(156) In FIG. **51** a pair of resistance members **142** are attached to an attachment point **526** positioned at a centre of the bow. The resistance members **142** sit in respective left and right hand channels **530** and the resistance members are attached to a resistance point **144** that is situated

behind the user when the device is in use.

(157) In FIG. 52 a pair of resistance members **142** is attached to respective attachment points **526** at opposite ends of the bow. In this arrangement the resistance members do not sit in the channel **530**, but extend directly from the attachment points to the resistance point **144**, situated behind the user when in use.

(158) Various configurations of the device can have one, two, or all three of the above described attachment points. The device can be supplied with one or two resistance members so that the user can use the device with one or two resistance members as preferred.

(159) In another example the device may be fitted with a pendulum or swinging member **600** or similar. The pendulum **600** is attached to the frame of the device at one end and is configured to freely swing in at least one plane and preferably is completely free to swing and not restricted to any particular direction. In the example shown in FIG. 53 the pendulum **600** is located at a front, centre point of the device when it is in use, and hangs below the lowest point of the frame. The movement of the device is shown more clearly in FIG. 54, which shows a view from the front of the device. The pendulum is free to swing side to side and will oscillate in response to a user's motion when the device is in use.

(160) The motion of the pendulum can be controlled by the user of the device when the device is in use. It is envisaged that the user can take a golf-like stance and use the device to improve their golf swing technique. Smooth twist movements from the user will allow for the pendulum to be controlled smoothly and its oscillation to be likewise controlled. Poor, jerky or mistimed technique will result in the oscillation of the pendulum being interrupted. The images provided in FIG. 55 show a device having a pendulum in use. The user can be seen to be in an approximate golf-like stance, with the device held at angle relative to an axis of his body. In image **1001** the user has twisted to the left (that is, his left) and the pendulum likewise swings to the left. The pendulum reaches its maximum displacement to the left and then begins to swing back towards the centre as seen in image **1002**, the user twists back towards the centre to control this movement. As the pendulum passes the centre point the user twists to the right and the pendulum reaches its maximum displacement to the right, as shown in image **1003**. This motion is then reversed, as the pendulum swings from right to left, as shown in images **1004-1006**.

(161) The pendulum may be positioned elsewhere and not necessarily on the frame. FIG. 56 provides an example of another arrangement, in which the pendulum is suspended from the resistance member when in use. In this use case the user can use the resistance member to exercise whilst the pendulum provides a cadence for the user to match. In this example, and the previous ones, the pendulum itself may be considered a resistance member. The mass of the resistance member may be configured to provide a weight against which the user can exercise. The pendulum may take a variety of forms, as well as being suspended below the device it is conceivable that a mass on a spring could be used above the plane of the device.

(162) The pendulum may comprise an elongate member. The elongate member may be rigid, or it may be flexible. The elongate member may have a mass at a location, for example its end, remote from where it is affixed to the frame of the device.

(163) The construction of the frame of the device is shown in FIG. 57. The top part of the frame has been removed, showing the internal construction. The frame is predominantly hollow, having multiple structural webs and and/or ribs that provide structural rigidity.

(164) FIG. 57 also shows a hinged example of the handles **514**. The handles are hinged inwards. This can allow users to vary their grip. In some embodiments, the handles **514** may be angled inwards towards the centre of the device **500** by means of a fixed angled joint between each handle **514** and the frame **510**. The hinges also provide a means for collapsing the handles for transport without having to remove them.

(165) Further detail regarding the resistance member attachment point is shown in FIGS. 58 to 60. FIG. 58 shows an attachment point at a centre point of the frame, situated within the channel. FIG.

59 shows an attachment point proximal to an end point of the frame, also situated within the channel. The attachment point comprises a column in the channel or defining part of the channel around which a clip—such as a carabiner—can be attached, for assembling the device with a resistance member; FIG. **60** shows a carabiner in place on the attachment point. There is a space behind the column into which the carabiner clip can extend so as to surround the column.

(166) FIG. **61** shows another variant of the device in which the inner ring or bow and the outer ring or bow have offset centres and in which the opening at the rear, that a user puts their waist/torso into, has a space **700** between the “horns” of the bow of about 61 cm+/-3 cm or +/-5 cm, or +/-7 cm or +/-10 cm.

(167) FIG. **62** shows another embodiment of the device **500**. The handles **614** of the device **500** each comprise two upright sections **615**. A first end of each upright section **615** is attached to the frame **510** of the device, for example as described above with reference to other embodiments. Each handle **614** further comprises a transverse section **616** which extends between the upright sections **615**. The angle between the transverse section **616** and each of the upright sections **615** is substantially a right angle; however, in other embodiments this angle may be less than or greater than 90 degrees.

(168) In use, a user may grip either upright section **615** of each handle **614**, or the transverse section **616** of each handle to vary their grip on the device **500**. In the embodiment of FIG. **62**, the device **500** comprises two handles **614**. In other embodiments, the device **500** may comprise fewer or more than two handles **614**. For example, the device **500** may comprise a single handle **614** such that in use, a user grips one of the upright sections **615** with each hand.

(169) In the embodiment of FIG. **62**, the each upright section **615** of each handle **614** extends substantially perpendicularly to the plane of the frame **510** of the device. In other embodiments, the each upright section **615** of each handle **614** may extend at an angle of less than or greater than 90 degrees to the plane of the frame **510**. For example, each upright section **615** of each handle **614** may be angled inwards towards the centre of the device **500**.

(170) The handle **614** of the embodiment of FIG. **62** may be used in place of the or each handle of any other embodiment described herein.

(171) Another example of an exercise device **600** is shown in FIG. **63**. The device **600** comprises a first frame section **116** and a plurality of second frame sections **118** mounted to the first frame section **116**. In this embodiment, the first and second frame sections **116**, **118** are formed from metal. A strap **128** is attached to the first frame section **118** so that the device **600** can be supported by a user **150** in use, as described above with reference to other embodiments. Each of the second frame sections **118** is substantially U-shaped and comprises a semi-. circular body extending from a first end to a second end. The first and second ends are mounted to the first frame member **116** by any suitable means, such as welding.

(172) The device **600** comprises a discontinuous radially outwardly facing channel. Each of the second frame sections **118** comprises part of the channel. Each part of the radially outwardly facing channel is configured to receive a portion of a resistance member **142**. In use, a first end of the resistance member **142** is attached to the device **600** and a second end of the resistance member **142** is attached to an external resistance point, as described above with reference to other embodiments. When the resistance member **142** is received in the channel in use, the resistance member **142** forms an arc which is tangential to one or more of the second frame sections **118**; FIG. **63** illustrates a use case where the arc of the resistance member **142** is tangential to two of the second frame sections **118**.

(173) In the embodiment of FIG. **63**, the device **600** comprises three second frame sections **118** spaced equidistantly about the circumference of the first frame section **116**. In other embodiments, the device **600** may comprise fewer or more than three second frame sections **118**.

(174) The device **600** further comprises a substantially U-shaped handle **714**. The handle **714** comprises a semi-circular body extending from a first end to a second end. The first and second

ends are mounted to the first frame section **116**. The handle **714** provides various grip positions and different points along the length of the body of the handle **714**.

(175) Another example of an exercise device **700** is shown in FIG. **64**. The exercise device **700** of FIG. **64** shares features in common with the exercise device **600** of FIG. **63**. Like reference numerals are used to refer to like features.

(176) The device **700** of FIG. **64** differs from the exercise device **600** of FIG. **63** in that the second frame members **118** are substantially linear shape. Each of the second frame members **118** comprises a first end and a second end distal from the first end. The first end of each of the second frame members **118** is mounted to the first frame member **116** by any suitable means, such as welding. The second end of each of the second frame members **118** comprises part of the discontinuous radially outwardly facing channel.

(177) Another example of an exercise device **800** is shown in FIGS. **65-70**. FIG. **65** shows a front view of the device **800**. The device **800** comprises a first frame member **116**, a second frame member **118** and two handles **814**. The second frame member **118** forms an arc of a nominal circle when viewed in plan. The central angle of the arc is greater than 180 degrees. The first frame member **116** extends across the second frame member **118** between two points spaced from the ends of the second frame member **118**. The first frame member may be arcuate or straight. When arcuate, it may have a radius of curvature that is greater than that of the first frame member. In use, a user rests the first frame member **116** against their torso and grips one of the handles **814** in each hand in a similar manner as described above with reference to other embodiments. In the embodiment of FIG. **65**, the device **800** comprises padding on the first frame member **116** to provide comfort to a user. In other embodiments the padding may not be present. The padding could be a foam tube, possibly with a longitudinal slit to allow it to be slipped on to the first frame member. The padding/foam tube could be replaceable/removable and replaceable, possibly for when it becomes worn.

(178) The device **800** further comprises an upper rail **815** and a lower rail **816**. The rails may form upper and lower lips of a discontinuous (or reduced depth) channel in which in use a cable, resilient rope or band or other or other resistance member is retained or held as the user twists their torso. The rails may provide a castellated series of projections and lands between the projections. The castellations of the upper and lower rail form radially outwardly flaring grooves or valleys in which the resistance member is received.

(179) The resistance member may be a bungee cable or the like, for example an elastic resistance band that exerts progressively more resistance as it is stretched, or an inelastic cable such as a metal cable or the like.

(180) FIG. **66** shows a close-up view of a portion of the device **800**. The upper rail **815** comprises a plurality of sections **817** arranged over the circumference of the second frame member **118**. Each section **817** comprises two first portions **8171** spaced apart from one another about the circumference of the second frame member **118**. Each first portion **8171** extends away from the second frame member **118** and extends at an acute angle to the plane of the circumference of the second frame member **118**. The first portions **8171** are also angled towards each other. Each section **817** further comprises a second portion **8172** extending between the ends of the first portions **8171** distal from the second frame member **118**. The lower rail **816** also comprises a plurality of sections **818** each comprising two first portions **8181** and a second portion **8182**. The sections **818** of the lower rail **816** are arranged in the same manner as the sections **817** of the upper rail **815**, but such that the first portions **8181** of the sections **818** of the lower rail **816** extend away from the first portions **8171** of the sections **817** of the upper rail **815**. The upper and lower rails **815**, **816** are arranged such that, in use, the first portions **8171** of each the section **817** of the upper rail **815** are angled away from the ground, and the first portions **8181** of each the section **818** of the lower rail **816** are angled towards the ground.

(181) The upper rail **815** is coupled to the second frame member **118** at the apex of the adjacent

first portions **8171** of adjacent sections **817** of the upper rail **815** by a joint **819**, such that the upper rail **815** protrudes from the second frame member **118** in a direction parallel to the plane of the second frame member **118**. The lower rail **826** is coupled to the second frame member **118** in the same manner. The joints **819** may be any suitable joint, such as a weld joint. The joints **819** of the upper and lower rails **815**, **816** are aligned with the same point along the circumference of the second frame member **118**. The joints **819** of the upper and lower rails **815**, **816** are spaced apart from each other in a direction perpendicular to the plane of the second frame member **118** to form a discontinuous (or periodically reduced depth) channel **830** extending about the circumference of the second frame member **118**. The channel **830** is configured to receive a resistance member. FIG. **66** shows a resistance member **142** received within the channel **830**. In use, the first and second portions **8171**, **8172** of the upper rail **815** and the first and second portions **8181**, **8182** of the lower rail **816** urge the resistance member **142** into the channel, thereby inhibiting the resistance member **142** from being dislodged from the channel **830**. The rails may be inclined at an angle to the central plane of the frame so as to subtend an angle of 35-70 degrees between them, or 45-60 degrees. (182) FIGS. **67** and **68** each show a further close-up view of the device **800**, showing how the joints **819** form the channel **830**. FIG. **67** shows the device **800** without a resistance member received within the channel **830** and FIG. **68** shows the device **800** with a resistance member **142** received within the channel **830**.

(183) FIG. **69** shows a rear view of the device **800**. In this embodiment, the device **800** is formed as a hollow box section of a suitable metal, such as steel, stainless steel or aluminium. The second frame member **118** may be formed by bending a straight box section of metal into the desired shape. The device **800** comprises an attachment point **820** for attaching an end of a resistance member to the device **800** in a similar manner as described above with reference to other embodiments. The rails may be welded to the frame member.

(184) FIG. **70** shows a plan view of the device **800** as viewed in a direction from the lower rail **816** towards the upper rail **815**. FIG. **70** shows a resistance member **142** attached to one of the attachment points **820**, by means of a carabiner **528**, and received within the channel **830**.

(185) The rails of the embodiments shown may be continuous pieces of material (eg metal rod) attached to the frame of the device at spaced apart regions.

(186) The rails may be welded to the outside surface of the frame, as shown in FIG. **67**. Also as shown in FIG. **67** the welding “blob” of metal used to weld the rail to the frame may be disposed on the side of the rail away from the channel that is to receive the cable/resistance member. This can leave the interior surfaces of the rails that face the channel and form the channel free of weld material, leaving a smoother channel.

(187) The radially outwardly facing wall of the frame that part-defines the channel and that in use contacts and receives the resistance member may have a concave curvature so as to assist in positioning/holding the resistance member. That concave curve may be formed naturally by the act of bending rectangular hollow section tube to form the curved frame. The flat metal wall on the outside of a bent hollow rectangular cross section metal (e.g. steel) tube may naturally have an outwardly facing concave curve or camber on the outer “flat” face.

(188) FIG. **71** shows a schematic plan view of the device **800** of FIGS. **65** to **70** being used by a user **150**. The user **150** holds the device **800** in a similar manner as described with reference to other embodiments of the invention. FIG. **71** shows a first end of a resistance member **142** attached to a section of the upper rail **815** by means of a carabiner **528**. In the configuration shown in FIG. **71**, the first end of the resistance member **142** is attached to a section of the upper rail **815** which is substantially directly in front of the user **150**. A second end of the resistance member **142**, opposite the first end, is attached to an external resistance point **144**. FIG. **71** shows the user **150** adopting a first position with respect to the external resistance point **144**. In the first position, the external resistance point **144** is substantially directly in front of the user **150**. As such, the carabiner **528** is located in the centre of the section of the upper rail **815** and the resistance member **142** extends

straight out in front of the user. FIG. 72 shows a schematic front view of a portion of the device **800** in the configuration shown in FIG. 71.

(189) FIG. 73 shows the same view as FIG. 71, but with the user **150** adopting a second position relative to the external resistance point **144**. In this position, the resistance member **142** exerts a clockwise torque on the device **800** and the user **150** resists this torque as a means of exercise, as described with reference to other embodiments of the invention. As the user **150** moves from the first position to the second position, the carabiner **528** slides along the section of the upper rail **815** until it reaches the end of the section of the upper rail **815** closest to the external resistance point **144**. Due to the angle of the upper rail **815** relative to the plane of the circumference of the second frame member **118**, this sliding of the carabiner **528** causes the resistance member **142** to extend through a gap between the section of the upper rail **815** and an adjacent section of the upper rail **815**, and become received within the channel **830**. FIG. 74 shows a schematic front view of a portion of the device **800** in the configuration shown in FIG. 73.

(190) FIG. 75 shows the same view as FIG. 71, but with the user **150** adopting a third position relative to the external resistance point **144**. In this position, the resistance member **142** exerts an anti-clockwise torque on the device **800** and the user **150** resists this torque as a means of exercise, as described with reference to other embodiments of the invention. The user **150** may move from the second position to the third position. As the user **150** does so, the carabiner **528** will slide from one end of the section of the upper rail **815** to the centre of the section of the upper rail **815**, and from the centre of the section of the upper rail **815** to the other end of the section of the upper rail **815**. Due to the angle of the upper rail **815** relative to the plane of the circumference of the second frame member **118**, this will cause the resistance member **142** to lift out of the channel **830** as the carabiner **528** slides to the centre of the section of the upper rail **815**, and back in to the channel **830** as the carabiner **528** slides to the other end of the section of the upper rail **815**. FIG. 76 shows a schematic front view of a portion of the device **800** in the configuration shown in FIG. 75.

(191) The arrangement of FIGS. 71 to 76 allows the user to transition between performing exercises with a clockwise torque exerted on the device **800** to performing exercises with an anti-clockwise torque exerted on the device **800** without having to detach and reattach the carabiner **528**. The user **150** is simply required to change their position relative to the external resistance point **144**. In addition, the sliding movement of the carabiner **528** along the section of the upper rail **815** as the user **150** changes position encourages the resistance member **142** in and out of the channel **830**, reducing or removing the effort required of the user **150** to ensure the resistance member **142** is properly received in the channel **830** before exercising.

(192) The castellated/intermittent/interrupted projections nature of the upper and lower rails, with projecting sections that are spaced from the frame member, and “gap” sections that are close to the frame member allows for gaps (viewed from above) between projection portions of the rails, and the resistance member can naturally use the gaps to feed into the groove defined by the upper and lower rails. By sliding to one end or the other, in use, of a castellated pair of upper and lower rail sections the resistance member is automatically placed adjacent the gaps in use at the point where it is connected to the selected rails. This makes threading the resistance member into the channel defined by the upper and lower rails easy for a user.

(193) It will be appreciated that in other embodiments, the resistance member **142** may be attached to a different section of the upper rail **815**, or to a section of the lower rail **816**. A user may attach the resistance member **142** to any section of the upper or lower rails **815**, **816** as appropriate.

(194) FIG. 77 shows a schematic plan view of another embodiment of the device **800**. In this embodiment, the device comprises a pulley **900**. A carabiner **528** (or other coupling for a resistance member) is attached to the pulley **900** and a resistance member **142**, and the pulley **900** is attached to a section of the upper rail **815**. In other embodiments, the pulley **900** may be attached to a different section of the upper rail **815** or to a section of the lower rail **816**. The pulley **900** is releasably attachable to any section of the upper or lower rails **815**, **816** to enable a user of the

device **800** to alter the position of the pulley **900**.

(195) In another variant the resistance member may be attached directly to one of the rails.

(196) FIG. **78** shows a schematic plan view of a portion of the device **800** comprising the pulley **900**. The pulley **900** comprises a housing **901** and a pulley wheel **902** rotatably mounted within the housing **901**. The pulley **900** further comprises an aperture **903** for attaching the carabiner **528** to the pulley **900**. When the pulley **900** is attached to the section of the upper rail **815**, or any other section of the upper or lower rail **815**, **816**, the section of rail sits between the pulley wheel **902** and the aperture **903**. When tension is provided in the resistance member **142**, the pulley wheel **902** engages the section of rail to aid sliding of the carabiner **528** from one end of the section of rail to the other end, as described with reference to FIGS. **71** to **76**.

(197) FIG. **79** shows a schematic side view of the pulley **900**. The housing **901** is open at one end and at the sides. The aperture **903** is located towards the open end of the housing **901** and is formed of an upper part and a lower part. The pulley wheel **902** is rotatably mounted between upper and lower parts of the housing **901**. The circumferential surface of the pulley wheel **902** is concave towards the axis of rotation 'A' of the pulley wheel **902**.

(198) FIG. **80** shows a schematic side view of a portion of the device **800** comprising the pulley **900** with the resistance member **142** in tension. The concave circumferential surface of the pulley wheel **902** is complementary to the engaged portion of the section of the upper rail **815**. This aids sliding of the carabiner **528** from one end of the section of rail to the other end, as described with reference to FIGS. **71** to **76**. To attach the resistance member **142** to the section of the upper rail **815**, a user first receives the section of rail through the open end of the pulley housing **901**. The user then threads the carabiner **528** through the aperture **903**, either with the resistance member **142** attached to the carabiner **528** or before attaching the resistance member **142** to the carabiner **528**. The carabiner **528** then prevents the pulley **900** from falling off the section of the upper rail **815** through the open end of the pulley housing **901**.

(199) FIG. **81** shows the device **800** in use. For clarity, not all of the features of the device **800** are labelled in FIG. **81**. In this example, each end of the second frame member **118** of the device **800** is attached to a resistance member **142** at a first end of the respective resistance member **142**. Each end of the second frame member **118** may be attached to the respective resistance member **142** by means of one of the attachment points **820** shown in FIG. **70** and a carabiner, for example. In other examples, each resistance member **142** may be attached to the second frame member **118** at any suitable location along the second frame member **118**.

(200) A second end of each resistance member **142**, opposite the respective first end, is attached to an external resistance point **144**. In this example, each external resistance point **144** comprises a bracket mounted to a wall by suitable means, such as one or more bolts. The user is facing away from the wall, with the device **800** in front them, and is gripping the handles of the device. The user can then rotate in either a clockwise or anticlockwise direction from the position shown to train the core muscles. In the example of FIG. **81**, there is no resistance member received in the radially outwardly facing channel of the device **800**. This demonstrates that the device **800** can optionally be utilised without a resistance member received in the radially outwardly facing channel. A similar use may involve a single resistance member attached at a first end of the resistance member to the second frame member **118** at any suitable location along the second frame member **118**, and a second end of the resistance member, opposite the first end, attached to a suitable external resistance point.

(201) The various features and characteristics of the examples and embodiments described within this specification are interchangeable between the various examples and embodiments. Any attachment means, adjustment means, methods of using the device, to name just a few examples, are not limited to the example with which they are associated in the description but may be added or substituted in to any of the other examples and embodiments of the exercise device.

Claims

1. An exercise device for exercising core muscles of a user's torso, the exercise device comprising: a frame, configured to be supported by the user; and a radially outwardly facing channel, wherein the channel is configured to receive an elongate resistance member, the channel comprising at least one of a concave outwardly facing channel and an outwardly facing flat sided and V-shaped channel; wherein the channel is configured in use to cause the elongate resistance member that is in use in the channel to be urged towards the centre of the channel by the shape of the channel; and wherein the frame defines an aperture configured to receive a torso of the user, and wherein the device is configured to provide a resistance force against which the user can exercise when in use.
2. The exercise device according to claim 1, wherein at least one of i) the frame is configured to be supported by the user such that in use the frame is supported by the forearms of the user with the user's forearms substantially parallel to the ground, and ii) the aperture is an open aperture, in that the frame does not completely surround a user when the device is in use, and wherein the frame extends in front of the user, at least partially to the sides of the user, and not behind the user when the device is in use.
3. The exercise device according to claim 1, wherein the frame comprises a maximum thickness measured substantially perpendicularly to a width of the frame, wherein a maximum width of the frame is at least two times the maximum thickness of the frame.
4. The exercise device according to claim 1, wherein the frame is formed from an outer curved member and an inner member that meet at junctions between the outer curved member and the inner member; and at least one of i) wherein the inner member is curved and comprises an arc with a radius of curvature greater than a radius of curvature of an arc of the outer curved member, and ii) wherein the inner member is flexible.
5. The exercise device according to claim 1, wherein the frame has at least one of a curved bow and crescent shape.
6. The exercise device according to claim 1, comprising at least one of i) a plurality of attachment points for attaching the resistance member to the frame, and ii) an attachment point for attaching the resistance member at a centre point of the frame.
7. The exercise device according to claim 1, wherein the channel is discontinuous.
8. The exercise device according to claim 7, wherein the frame comprises a first frame section and a plurality of second frame sections extending radially outwardly from the first frame section, wherein each of the second frame sections comprises part of the channel.
9. The exercise device according to claim 7 comprising an upper rail and a lower rail, wherein each of the upper rail and lower rail are coupled to the frame by a joint at a plurality of points along a length of the frame, wherein the rails and the joints form the discontinuous channel.
10. The exercise device according to claim 9, wherein at least one of i) one or more of the plurality of joints coupling the upper rail to the frame is aligned along a length of the frame with one or the plurality of joints coupling the lower rail to the frame, and ii) each of the upper rail and the lower rail comprises a plurality of sections, wherein each section extends from the frame and each joint is located at an apex between adjacent sections.
11. The exercise device according to claim 9, wherein the resistance member is attachable to at least one of the upper rail and the lower rail, and, wherein the exercise device comprises a pulley for attaching the resistance member to at least one of the upper rail and the lower rail, wherein the pulley comprises a pulley wheel rotatably mounted within a housing and an aperture for attaching the resistance member to the pulley, wherein the rail sits between the pulley wheel and the aperture of the pulley when the resistance member is attached to the rail via the pulley.
12. The exercise device according to claim 1, wherein the frame comprises a box section, wherein the box section is formed from metal.

13. The exercise device according to claim 1, comprising at least one of (i) an indicator to provide a user of the device with information relating to their use of the device; (ii) a mount for mounting an electronic device to the exercise device; and (iii) an attachment for attaching an electronic device to the exercise device.
14. The exercise device according to claim 1, wherein the frame lies on a plane and the channel is extended at one or more points in a direction having a component transverse to the plane, and wherein the extension or extensions to the channel comprise one or more shields to protect a user of the device in the event of accidental disconnection of the resistance member, wherein the one or more shields are releasably attachable to the frame.
15. The exercise device according to claim 1, further comprising handles for providing a device supporting means to the user, and wherein the handles are at least one of (i) multiple sets of handles to provide differing hand/arm positions for the user when the device is in use, (ii) adjustable relative to the frame to provide differing hand/arm positions for the user when the device is in use, and (iii) adjustable relative to the frame, wherein the frame defines one or more slots configured to receive the handles and the handles are angularly moveable within the slots, wherein the handles are securable at a plurality of positions within the slots.
16. The exercise device according to claim 15, wherein at least one of: (i) the handles are pivotable relative to the frame, (ii) each handle comprises two upright sections and a transverse section extending between the upright sections, and (iii) the frame is formed from an outer curved member and an inner member that meet at junctions between the outer curved member and the inner member, wherein the handles are attachable to the outer curved member.
17. The exercise device according to claim 1, wherein the aperture is partly defined by the frame and partly defined by another component, wherein the other component comprises at least one of a strap and a support, and wherein the exercise device comprises a lumbar support attached to the at least one of a strap and support or formed within the at least one of a strap and support, and wherein the frame has a curved bow or crescent shape and the at least one of a strap and support extends in use from the ends of the bow or crescent shaped frame.
18. The exercise device according to claim 1, further comprising a plurality of resistance members attached to the frame.
19. A kit of parts comprising: the exercise device of claim 1 and a resistance member; and a set of instructions for providing a user with methods of using the device; and a stability training device.
20. An exercise device for exercising core muscles or a user's torso, the exercise device comprising: a frame, configured to be supported by the user; an elongate resistance member; and a radially outwardly facing channel, the channel having a depth of at least four times the diameter of the elongate resistance member; wherein the channel is configured to receive the elongate resistance member, the channel comprising at least one of a concave outwardly facing channel and an outwardly facing flat sided and V-shaped channel; wherein the channel is configured in use to cause the elongate resistance member that is in use in the channel to be urged towards the centre of the channel by the shape of the channel; wherein the frame defines an aperture configured to receive a torso of the user, wherein the frame further comprises a first frame section and a plurality of second frame sections extending radially outwardly from the first frame section, wherein each of the second frame sections comprises part of the channel, and wherein the device is configured to provide a resistance force against which the user can exercise when in use; and wherein the channel extends between the handles such that, in use, at least a portion of the channel is positioned in front of the user.
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